



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

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Mid Semester Examination

March 2018

Max. Marks: 30

Class: FYMCA :II

Course Code: MCA21 : MCA

Name of the Course: Operating System

Duration: 1.5 Hrs

Instruction:

- (1) All questions are compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Q No.	Question	Max. Marks	CO																					
Q.1	Classify different OS based on their functions OR Illustrate types of system calls	05	CO1 CO1																					
Q.2	Demonstrate process state diagram.	05	CO2																					
Q.3	Cosider the following set of processes. Draw Gantt chart,Find average waiting time,average turnaround time for Round Robin scheduling algorithm(Time Quantum=03) <table border="1"><thead><tr><th>Process No</th><th>Arrival Time</th><th>Burst Time</th></tr></thead><tbody><tr><td>1</td><td>5</td><td>5</td></tr><tr><td>2</td><td>4</td><td>6</td></tr><tr><td>3</td><td>3</td><td>7</td></tr><tr><td>4</td><td>1</td><td>9</td></tr><tr><td>5</td><td>2</td><td>2</td></tr><tr><td>6</td><td>6</td><td>3</td></tr></tbody></table>	Process No	Arrival Time	Burst Time	1	5	5	2	4	6	3	3	7	4	1	9	5	2	2	6	6	3	05	CO2
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6	6	3																						
Q 4.	There is a buffer of 5 slots and each slot is capable of storing one unit of data. There are two processes running, namely, producer and consumer, which are operating on the buffer. producer produces an article and consumer consumes it.How does this problem can be solved using semaphores	05	CO2																					

Q.5

Consider the system with 4 processes P₀, P₁, P₂, P₃ and three resources of type A, B, C. The following snapshot is given

05

CO2

	Max	Allocation	Available
	A B C	A B C	A B C
P ₀	0 0 1	0 0 1	
P ₁	1 7 5	1 0 0	
P ₂	2 3 5	1 3 5	
P ₃	0 6 5	0 6 3	
Total		2 9 9	1 5 2

Using Banker's algorithm

1. What is the content of Need matrix?

2. Is the system in safe state? If so, give safe sequence

OR

Analyze whether given task sets are schedulable or not using Rate Monotonic Scheduling algorithm. (Assume 1 is the highest priority)

CO2

Process No	Priority	Time Period(Repeat Frequency)	Service Time
1	3	20	3
2	1	5	2
3	2	10	2

Q.6

Consider following page reference string

1 2 3 2 1 5 2 1 6 2 5 6 3 1 3 6 1 2 4 3

How many page faults would occur for Optimal page replacement algorithm, assuming 3 frames. initially all frames are empty

05

CO3